

Introduction to abelian categories

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Abstract

We develop the theory of abelian categories, motivated by homological algebra in $\mathbf{Vect}_k$. We then develop the notions of homology, exactness, derived functors and chain homotopy.

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1 Introduction

Throughout the Colloquium (I hope I spelled that right) we have discussed the theory of homology and its applications at length, but we have always taken our chain complexes to have values in vector spaces, or more generally R -modules. A natural question is whether we can construct chain complexes in more general categories, broadening our class of objects to which we can associate a theory of homological algebra.

However, homological algebra has a large number of requirements, both obvious and non-obvious. For example, given a chain complex $C_\bullet$, the n th homology $H_n(C_\bullet)$ is the quotient $\text{Im } d_{n-1} / \text{Ker } d_n$. Even just to construct homology, we need to define kernels, images, and some kind of quotient. Kernels don't even exist in many mundane categories, such as $\mathbf{Set}$ and $\mathbf{Top}$. Additionally, it may not be clear how to abstract images and kernels to non-concrete categories, since they are fundamentally constructed using the underlying sets of objects.

To this end, we develop the theory of *abelian categories* as an axiomatization of homological algebra. The theory of abelian categories is attributed to Grothendieck in [Gro57] and was developed for this very purpose, and all constructions in homological algebra can be traced back to the theory of abelian categories in some way or another.

Throughout this talk, we will develop the theory of abelian categories using this motivation. We will then construct homology in these categories, and introduce some of the basic theory of diagram chasing. If time permits, we will close with a discussion of derived functors.

Convention 1.1. Given a category $\mathcal{C}$, we write $x, y \in \mathcal{C}$ to denote that x, y are objects in $\mathcal{C}$, and $\mathcal{C}(x, y)$ to denote the hom-set of morphisms $x \rightarrow y$ in $\mathcal{C}$. Objects in arbitrary categories will usually be denoted by lowercase letters. Important categories are denoted with capital letters in script and lowercase letters in roman (eg. $\mathbf{Top}, \mathbf{Set}, \mathbf{Ab}$). Functors are usually denoted by capital roman

letters $F : \mathcal{C} \rightarrow \mathcal{D}$, unless there are other conventions (eg. quotient functors denoted by π). We will always denote the group operation in abelian groups by addition, and will always denote the group operation in non-abelian groups by multiplication. The letter k always denotes a field, the letter R always denotes a ring, and the categories of k -vector spaces and R -modules are denoted Vect_k and Mod_R , respectively.

As the reader can likely expect, this talk will lean somewhat heavily into category theory. We will assume the familiar is familiar with the definition of a category, a functor, hom-sets, and all other required concepts will be introduced in the main text. We also offer a refresher on the basic definitions in ??.

2 Ab-categories

Let $\mathcal{C}$ be a category. In order to do homological algebra in $\mathcal{C}$, we need to be able to at the very least construct *chain complexes* and *homology*. We will begin by constructing chain complexes, which are considerably simpler.

We begin by reviewing the definition of chain complexes in the category Vect_k of k -vector spaces; the reader unfamiliar with vector spaces over an arbitrary field k can take all vector spaces to be over $\mathbb{R}$. A *chain complex* $(V_\bullet, d_\bullet)$ is a collection $\{V_i\}_{i \in \mathbb{Z}}$ of k -vector spaces, and a collection $\{d_i\}_{i \in \mathbb{Z}}$ of k -linear maps known as the *boundary maps* or *differentials*, as depicted in the diagram

$$\cdots \xrightarrow{d_{n+2}} V_{n+1} \xrightarrow{d_{n+1}} V_n \xrightarrow{d_n} V_{n-1} \xrightarrow{d_{n-1}} \cdots,$$

such that $d_{i-1} \circ d_i = 0$ for all i . We would like to elevate this definition to an arbitrary category $\mathcal{C}$. Luckily, the definition is already largely categorical, since we can simply take each V_i to be an object of $\mathcal{C}$, and each d_i to be a morphism $V_i \rightarrow V_{i-1}$ in $\mathcal{C}$. The main issue is the condition that $d_{i-1} \circ d_i = 0$. In Vect_k , the zero-morphism $0 : V \rightarrow W$ is the map which sends each $v \in V$ to the additive identity $0 \in W$. Unfortunately, objects in arbitrary categories don't have additive identities nor do they have elements, so we need to find a new characterization of the zero morphism which is more categorical. The way we do this is by recalling that the hom-set $\text{Vect}_k(V, W)$ is an *abelian group*.

Definition 2.1. An *abelian group* is a set G together with an operation $+$ such that

1. (closure) For all $g, h \in G$, we have $g + h \in G$;
2. (associativity) For all $g, h, k \in G$,

$$(g + h) + k = g + (h + k);$$

3. (identity) There exists an element $0 \in G$ such that for all $g \in G$,

$$0 + g = g + 0 = g;$$

4. (inverse) For all $g \in G$, there is some element $-g \in G$ such that

$$g + (-g) = (-g) + g = 0;$$

5. (commutativity) For all $g, h \in G$,

$$g + h = h + g.$$

Remark 2.2. An abelian group can be thought of as a vector space where the only scalars are the *integers*, and scalar multiplication is defined as follows. If A is an abelian group, $a \in A$, and $n \in \mathbb{Z}$, then $n \cdot a$ is simply defined as $\underbrace{a + \cdots + a}_{n \text{ times}}$.

Proposition 2.3. Let V, W be k -vector spaces. Then $\text{Vect}_k(V, W)$ is an abelian group. In particular, if $\varphi, \psi : V \rightarrow W$ are k -linear maps, then $\varphi + \psi$ is the k -linear map defined by

$$(\varphi + \psi)(v) = \varphi(v) + \psi(v),$$

and the additive identity is the zero map $0 : V \rightarrow W$ given by $0(v) = 0$.

The proof of proposition 2.3 is not difficult, and an even stronger result is proven in basic linear algebra courses, which is that $\text{Vect}_k(V, W)$ is a k -vector space. However, the result tells us something quite interesting: in an arbitrary category $\mathcal{C}$, if the hom-sets $\mathcal{C}(x, y)$ have an abelian group structure, then we can define the zero morphism as the additive identity of this set. This motivates definition 2.4:

Definition 2.4. Let $\mathcal{C}$ be a category. We say $\mathcal{C}$ is an *Ab-category* if for all $x, y \in \mathcal{C}$, the hom-set $\mathcal{C}(x, y)$ has an *abelian group structure* (we always denote the group operation by $+$), and if composition distributes over addition. Concretely, this means that if $f, g : x \rightarrow y$ and $h, k : y \rightarrow z$ are morphisms in $\mathcal{C}$, then

$$h \circ (f + g) = (h \circ f) + (h \circ g) \quad \text{and} \quad (h + k) \circ f = (h \circ f) + (k \circ f).$$

To see why we require that composition distribute over multiplication, recall that elements of $\text{Vect}_k(V, W)$ can be represented by matrices (in the finite dimensional case), and function composition is simply matrix multiplication. We then simply note that addition distributes over multiplication. We will see stronger motivation for this requirement later.

Remark 2.5. If the reader is familiar with enriched category theory, they will recognize that an *Ab-category* in the sense of definition 2.4 is precisely the underlying category of an $(\text{Ab}, \otimes)$ -enriched category, where we impose the group structure by identifying along the natural isomorphism $\text{Ab}(\mathbb{Z}, -) \cong U$, for $U : \text{Ab} \rightarrow \text{Set}$ the forgetful functor.

Notation 2.6. Let $\mathcal{C}$ be an *Ab-category*. We denote the additive identity of $\mathcal{C}(x, y)$ by 0 , and will call it the zero morphism. We will often not specify the domain and codomain of a zero morphism, and will leave it implicit.

Definition 2.7. Let $\mathcal{C}$ be an *Ab-category*. A *chain complex* $(c_\bullet, d_\bullet)$ in an *Ab-category* is a collection $\{c_i\}_{i \in \mathbb{Z}}$ of objects of $\mathcal{C}$, and a collection $\{d_i\}_{i \in \mathbb{Z}}$ of morphisms of $\mathcal{C}$

$$\cdots \xrightarrow{d_{n+2}} c_{n+1} \xrightarrow{d_{n+1}} c_n \xrightarrow{d_n} c_{n-1} \xrightarrow{d_{n-1}} \cdots$$

such that $d_{i-1} \circ d_i = 0$ for all i . The morphisms $d_\bullet$ are called *differentials* or *boundary morphisms*. We will usually denote a chain complex by $c_\bullet$, and will leave the notation for the boundary morphisms implicit.

Remark 2.8. A *cochain complex* is the dual notion of a chain complex. Concretely, given an *Ab-category* $\mathcal{C}$, a cochain complex $(c^\bullet, d^\bullet)$ is a collection of objects $\{c^i\}_{i \in \mathbb{Z}}$ and morphisms $\{d^i\}_{i \in \mathbb{Z}}$

$$\cdots \xrightarrow{d^{n-1}} c^{n-1} \xrightarrow{d^n} c^n \xrightarrow{d^{n+1}} c^{n+1} \xrightarrow{d^{n+2}} \cdots$$

such that $d^i \circ d^{i-1} = 0$ for all i . The homology of a cochain complex is called *cohomology*.

The astute reader has likely realized that the only difference between a chain and cochain complex is that we write the indices of a cochain complex as superscripts, and that they ascend rather than descend. Indeed, the theory of chain and cochain complexes is precisely the same, and the main reason we make a distinction between the two is that in many examples in nature, it is more natural to write the indices as either ascending or descending. For example, our objects could be sets of n -dimensional objects, and our boundary morphisms could take n -dimensional objects and produce $(n+1)$ -dimensional objects. It would make sense to set c^n to be the n -dimensional objects, and set $d^i : c^i \rightarrow c^{i+1}$, hence giving us a cochain complex.

Convention 2.9. It is also common practice to write the boundary morphisms of a (co)chain complex as ∂ rather than d . We stick to the convention of writing d for the boundary morphisms in this section.

3 Additive categories

The next step up from an $\mathcal{Ab}$ -category is an *additive category*. An extremely important object in homological algebra is the trivial k -vector space 0 . The trivial k -vector space is used as the “start” and “end” of many chain complexes, including short exact sequences, and also characterizes when there is no homology in a complex. To have a good theory of homological algebra, we should require that our categories have some kind of “zero object”. Luckily, regular category theory already provides a good definition.

Definition 3.1. A *zero object* or *null object* of a category $\mathcal{C}$ is an object $0 \in \mathcal{C}$ which is both initial and terminal.

Indeed, $0 \in \mathbf{Vect}_k$ is both initial and terminal. Additionally, given $V \in \mathcal{C}$ an $\mathcal{Ab}$ -category with a zero object, the unique maps $0 \rightarrow V$ and $V \rightarrow 0$ must be the zero map since abelian groups must have an identity. This is once again the same situation that we have in $\mathbf{Vect}_k$.

The next definition we introduce is largely motivated by $\mathbf{Vect}_k$. Many constructions and proofs in homological algebra require the following criteria, but we will unfortunately not see any in these notes. To motivate this notion, we present a fascinating theorem.

Proposition 3.2. Let $\mathcal{C}$ be an $\mathcal{Ab}$ -category, and $x, y \in \mathcal{C}$. If a product $x \times y$ of x, y exists, then a coproduct $x \amalg y$ also exists, and there is an isomorphism $x \times y \cong x \amalg y$. More generally, finite products and coproducts coincide. Dually, if $x \amalg y$ exists, then so does $x \times y$, and $x \times y \cong x \amalg y$.

The proof requires a simple lemma.

Lemma 3.3. Let $\mathcal{C}$ be an $\mathcal{Ab}$ -category, $x, y \in \mathcal{C}$, and $f : x \rightarrow y$. Then $f \circ 0 = 0$ and $0 \circ f = 0$ for all zero morphisms 0 .

Proof. By definition 2.4, composition distributes over addition, and by definition 2.1 we have $0 = 0 + 0$. We then have

$$f \circ 0 = f \circ (0 + 0) = (f \circ 0) + (f \circ 0).$$

We then subtract $f \circ 0$ from both sides, and note that $(f \circ 0) - (f \circ 0) = 0$ since they cancel. This yields

$$0 = f \circ 0.$$

The other statement follows in the same way. ■

Proof of proposition 3.2. We first consider the case of the empty (co)product. Let $*$ $\in \mathcal{C}$ be terminal. Then note that the only morphism $*$ $\rightarrow *$ must be 0 since $\mathcal{C}(*, *)$ is a singleton. Now let $c \in \mathcal{C}$, and let $f : * \rightarrow c$. At least one morphism must always exist, since $\mathcal{C}(*, c)$ is an abelian group, and the empty set is not a group. Then $f = f \circ 1_* = f \circ 0 = 0$, where the final step follows by lemma 3.3. Thus we have $f = 0$ for all $f : * \rightarrow c$, meaning that $\mathcal{C}(*, c)$ is a singleton. Hence f is initial. This proves the statement for the empty product and coproduct.

Now let $x, y \in \mathcal{C}$ have a product $x \times y$ in $\mathcal{C}$. We establish some important compatibilities of the $\mathcal{A}b$ -structure with the product. Given a diagram of solid arrows

$$\begin{array}{ccccc} & & x & & \\ & \nearrow f & \uparrow \pi_x & & \\ z & \xrightarrow{(f,g)} & x \times y & & \\ & \searrow g & \downarrow \pi_y & & \\ & & y & & \end{array}$$

we always denote the unique arrow, as indicated by the dashed arrow, by (f, g) . By commutativity of the diagram, $\pi_x \circ (f, g) = f$ and $\pi_y \circ (f, g) = g$. In particular, it is the unique morphism with this property.

Let $z \in \mathcal{C}$, and let $f_1, f_2 : z \rightarrow x$ and $g_1, g_2 : z \rightarrow y$. We claim

$$(f_1, g_1) + (f_2, g_2) = (f_1 + f_2, g_1 + g_2). \quad (\star)$$

Indeed, note that since composition distributes over addition,

$$\pi_x \circ ((f_1, g_1) + (f_2, g_2)) = (\pi_x \circ (f_1, g_1)) + (\pi_x \circ (f_2, g_2)) = f_1 + f_2,$$

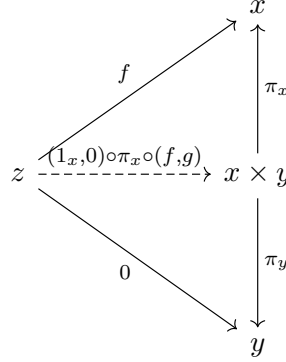
$$\pi_y \circ ((f_1, g_1) + (f_2, g_2)) = (\pi_y \circ (f_1, g_1)) + (\pi_y \circ (f_2, g_2)) = g_1 + g_2.$$

The claim follows by uniqueness of the relevant arrow.

We also claim that $(1_x, 0) \circ \pi_x + (0, 1_y) \circ \pi_y = 1_{x \times y}$. Indeed, let $f : z \rightarrow x$ and $g : z \rightarrow y$ be any morphisms. Then consider the diagram

$$\begin{array}{ccccc} & & x & & \\ & \nearrow 1_x & \uparrow \pi_x & & \\ & x & \xrightarrow{(1_x, 0)} & x \times y & \\ & \searrow 0 & \downarrow \pi_y & & \\ & & y & & \\ & \nearrow f & \uparrow \pi_x & & \\ z & \xrightarrow{(f,g)} & x \times y & & \\ & \searrow g & \downarrow \pi_y & & \\ & & y & & \end{array}$$

Noting that $f \circ 1_x = f$ and $0 \circ \pi_x \circ (f, g) = 0$ by lemma 3.3, we see that this diagram witnesses the commutativity of the product diagram



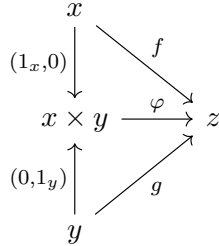
In particular, we see the equality $(1_x, 0) \circ \pi_x \circ (f, g) = (f, 0)$. The same logic yields $(0, 1_y) \circ \pi_y \circ (f, g) = (0, g)$. Recall by $(\star)$ that $(f, 0) + (0, g) = (f, g)$. This implies

$$\begin{aligned} (f, g) &= ((1_x, 0) \circ \pi_x \circ (f, g)) + ((0, 1_y) \circ \pi_y \circ (f, g)) \\ &= ((1_x, 0) \circ \pi_x + (0, 1_y) \circ \pi_y) \circ (f, g). \end{aligned}$$

In particular, postcomposition $((1_x, 0) \circ \pi_x + (0, 1_y) \circ \pi_y)_*$ is the identity map on $\mathcal{C}(x \times y, -)$. Since the Yoneda embedding is fully faithful, it reflects identities, and thus

$$(1_x, 0) \circ \pi_x + (0, 1_y) \circ \pi_y = 1_{x \times y}. \quad (\star\star)$$

The identity $(\star\star)$ highlights the importance of the arrows $(1_x, 0)$ and $(0, 1_y)$. Note that these define inclusions $x \rightarrow x \times y$ and $y \rightarrow x \times y$. We will now see that they yield a coproduct structure on $x \times y$. Let $f : x \rightarrow z$ and $g : y \rightarrow z$. Define $\varphi := f \circ \pi_x + g \circ \pi_y$. We claim that φ renders the diagram commutative:



This is a simple computation:

$$\begin{aligned} \varphi \circ (1_x, 0) &= (f \circ \pi_x + g \circ \pi_y) \circ (1_x, 0) \\ &= f \circ \pi_x \circ (1_x, 0) + g \circ \pi_y \circ (1_x, 0) \\ &= f \circ 1_x + 0 \\ &= f, \\ \varphi \circ (0, 1_y) &= (f \circ \pi_x + g \circ \pi_y) \circ (0, 1_y) \\ &= f \circ \pi_x \circ (0, 1_y) + g \circ \pi_y \circ (0, 1_y) \\ &= 0 + g \circ 1_y \\ &= g. \end{aligned}$$

Uniqueness follows by the identity eq. (★★). Suppose there exists $\psi : x \times y \rightarrow z$ such that the diagram

$$\begin{array}{ccc}
 x & & \\
 (1_x, 0) \downarrow & \searrow f & \\
 x \times y & \xrightarrow{\psi} & z \\
 (0, 1_y) \uparrow & \nearrow g & \\
 y & &
 \end{array}$$

commutes. Then

$$\begin{aligned}
 \varphi &= \varphi \circ 1_{x \times y} \\
 &= \varphi \circ ((1_x, 0) \circ \pi_x + (0, 1_y) \circ \pi_y) \\
 &= \varphi \circ (1_x, 0) \circ \pi_x + \varphi \circ (0, 1_y) \circ \pi_y \\
 &= f \circ \pi_x + g \circ \pi_y \\
 &= \psi \circ (1_x, 0) \circ \pi_x + \psi \circ (0, 1_y) \circ \pi_y \\
 &= \psi \circ ((1_x, 0) \circ \pi_x + (0, 1_y) \circ \pi_y) \\
 &= \psi \circ 1_{x \times y} \\
 &= \psi.
 \end{aligned}$$

Thus $\varphi = \psi$ is unique, and $x \times y = x \amalg y$ is a coproduct. ■

Definition 3.4. Let $\mathcal{C}$ be a category. Two objects $x, y \in \mathcal{C}$ have a *biproduct* when their product and coproduct coincide. We then call both the product and coproduct a *biproduct* and denote it by $x \oplus y$.

In practice, we rarely actually call the biproduct a biproduct, and simply call it a product or coproduct when appropriate. We also rarely use the notation $\oplus$ outside of *additive categories*, which we will shortly introduce.

The category $\mathbf{Vect}_k$ has finite products and coproducts, and by proposition 3.2 they coincide. The product of two k -vector spaces $V \times W$ is the same as their coproduct $V \amalg W$, so we call it a *direct sum* and denote it by $V \oplus W$. Direct sums are extremely useful in the theory of k -vector spaces, and more generally the theory of R -modules, so it would be nice if our $\mathcal{A}b$ -categories had finite biproducts as well. This motivates definition 3.5.

Definition 3.5. An $\mathcal{A}b$ -category $\mathcal{C}$ is an *additive category* if it has a zero object and binary biproducts. Equivalently, it must have a zero object and binary products (or coproducts), since products *are* coproducts in $\mathcal{A}b$ -categories by proposition 3.2.

4 Abelian categories

There is one problem we have not yet solved. In order to do homological algebra, we need to be able to compute *homology*, hence the name of the field. Our goal now will be to figure out what else an additive category needs in order to have a notion of homology. To begin, we recall the definition of homology in $\mathbf{Vect}_k$.

Let $(V_\bullet, d_\bullet)$ be a chain complex in Vect_k . Since $\text{Im } d_{i+1} \subseteq \text{Ker } d_i \subseteq V_i$, we defined the i th homology as

$$H_i(V_\bullet) = \frac{\text{Ker } d_i}{\text{Im } d_{i+1}}.$$

To define homology in an arbitrary category, we will thus need *kernels*, *images*, and *quotients*. We will begin with kernels, since they will motivate both of the other constructions.

Let $\varphi : V \rightarrow W$ be a k -linear map. The *kernel* $\text{Ker } \varphi$ is defined as the set of all $v \in V$ such that $\varphi(v) = 0$. In other words, it is the fiber of 0 under φ . We need to promote this to a diagrammatic definition that can be done in any additive category, or more generally, any $\mathcal{A}b$ -category. To do this, we make an important observation.

Remark 4.1. The kernel $\text{Ker } \varphi$ is a subspace of V , meaning there is an *inclusion map* $i : \text{Ker } \varphi \hookrightarrow V$ which maps each $v \in \text{Ker } \varphi$ to itself $v \in V$. This inclusion map is

1. a k -linear map,
2. a *monomorphism*,
3. and the composite $\varphi \circ i = 0$ because the image of i is precisely $\text{Ker } \varphi$, and $\varphi(\text{Ker } \varphi) = 0$.

We will explain monomorphisms shortly. For now, we illustrate another fact. Suppose we have some other k -linear map $\psi : Z \rightarrow V$ such that $\varphi \circ \psi = 0$. This implies that $\text{Im } \psi \subseteq \text{Ker } \varphi$, so we note that ψ can be decomposed as

$$\begin{array}{ccccccc} & & \psi & & & & \\ & \nearrow & & \searrow & & & \\ Z & \xrightarrow{\psi} & \text{Im } \psi & \hookrightarrow & \text{Ker } \varphi & \xrightarrow{i} & V \end{array}$$

for $\text{Im } \psi \hookrightarrow \text{Ker } \varphi$ the inclusion map. In particular, the entire situation can be displayed as the commutative diagram

$$\begin{array}{ccccc} & & 0 & & \\ & \nearrow & & \searrow & \\ Z & \xrightarrow{\psi} & V & \xrightarrow{\varphi} & W \\ & \searrow \psi & \uparrow i & \nearrow 0 & \\ & & \text{Ker } \varphi & & \end{array}$$

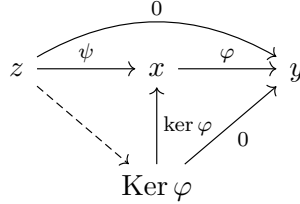
Here we have merged the maps $Z \rightarrow \text{Im } \psi$ and $\text{Im } \psi \hookrightarrow \text{Ker } \varphi$ into a single map $Z \rightarrow \text{Ker } \varphi$ for simplicity, which we also denote by ψ in a mild abuse of notation. The most interesting thing is that the map $\psi : Z \rightarrow \text{Ker } \varphi$ is *unique*; this is the *only* map that will make the diagram commute.

Anyone who has seen *universal properties* before will realize the significance of this. A classic fact of category theory (specialized to the case of kernels) is as follows. Suppose we have some other k -vector space K such that for any diagram of solid arrows

$$\begin{array}{ccccc} & & 0 & & \\ & \nearrow & & \searrow & \\ Z & \xrightarrow{\psi} & V & \xrightarrow{\varphi} & W \\ & \searrow \psi & \uparrow i & \nearrow 0 & \\ & & K & & \end{array}$$

there is a unique dashed arrow as indicated, rendering the diagram commutative. Then we necessarily have an isomorphism $K \cong \text{Ker } \varphi$. In other words, any other k -vector space satisfying this diagrammatic property must necessarily be isomorphic to the kernel. Since we only really care about objects up to isomorphism, we could choose to *define* the kernel in an arbitrary category as any k -vector space satisfying this property. Indeed,

Definition 4.2. Let $\mathcal{C}$ be an $\mathcal{Ab}$ -category, and let $\varphi : x \rightarrow y$ be a morphism in $\mathcal{C}$. A *kernel* of φ , denoted $\text{Ker } \varphi$, is an object $\text{Ker } \varphi \in \mathcal{C}$ together with a morphism $\text{ker } \varphi : \text{Ker } \varphi \rightarrow x$ such that for any diagram of solid arrows

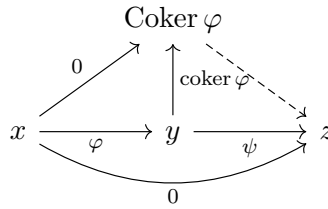


there is a unique dashed arrow as indicated, rendering the diagram commutative.

Note the shift in terminology, where we are now referring to the inclusion $\text{Ker } \varphi \hookrightarrow x$ as the *kernel*. We do this since morphisms are arguably more important than objects in arbitrary categories, so it is often more convenient to refer to the *morphism* as the kernel, rather than the object. We differentiate between the object and morphism by using a capital letter for the object.

We now have kernels in any $\mathcal{Ab}$ -category, so we only need to define images and quotients. Both of them will actually follow very easily from the notion of a kernel. First, we note that kernels have a dual notion given by reversing all the arrows in definition 4.2:

Definition 4.3. Let $\mathcal{C}$ be an $\mathcal{Ab}$ -category, and let $\varphi : x \rightarrow y$ be a morphism in $\mathcal{C}$. A *cokernel* of φ , denoted $\text{Coker } \varphi$, is an object $\text{Coker } \varphi \in \mathcal{C}$ together with a morphism $\text{coker } \varphi : y \rightarrow \text{Coker } \varphi$, such that for any diagram of solid arrows



there is a unique dashed arrow as indicated, rendering the diagram commutative.

Similarly to kernels, cokernels are also unique up to isomorphism. To see why this is a useful definition, it will be useful to explicitly construct cokernels in Vect_k .

Proposition 4.4. Let $\varphi : V \rightarrow W$ be a k -linear map. Then the cokernel can be constructed as the quotient $\text{Coker } \varphi \cong W / \text{Im } \varphi$.

Proof. The quotient $X \rightarrow X/Y$ is initial with respect to identifying Y with 0. The cokernel is initial with respect to identifying $\text{Im } \varphi$ to 0. The statement follows. $\blacksquare$

Remark 4.5. The definition of a quotient is inherently dependent on subsets, which do not exist in general in $\mathcal{Ab}$ -categories. However, proposition 4.4 allows us to *almost* recharacterize quotients using *cokernels*, which we have already defined in any $\mathcal{Ab}$ -category. Note that in Vect_k , if we have a subspace $W \subseteq V$, then there is an inclusion $i : W \hookrightarrow V$ with $\text{Im } i \cong W$. By proposition 4.4,

the cokernel $\text{Coker } i$ can be realized as the quotient $V/\text{Im } i$, but since $\text{Im } i \cong W$, we have that $\text{Coker } i \cong V/W$. This allows us to quotient by any subobject using cokernels. The issue then becomes that we do not have a notion of *subobjects* in general categories.

We once again turn to the theory of k -vector spaces. In Vect_k , every subspace is a kernel of some morphism. In particular, it is the kernel of its *cokernel*. We can actually prove something more general, which will be extremely useful.

Proposition 4.6. *Let $\varphi : V \rightarrow W$ be a k -linear map. Then the image $\text{Im } \varphi$ is given by the kernel of the cokernel $\text{Im } \varphi \cong \text{Ker}(\text{coker } \varphi)$.*

Proof. The cokernel is the map $W \rightarrow W/\text{Im } \varphi$ given by $w \mapsto w + \text{Im } \varphi$, using coset notation. By our knowledge of cosets from standard group theory, the kernel of this map is precisely $\text{Im } \varphi \subseteq W$. Thus, $\text{Ker}(\text{coker } \varphi) \cong \text{Im } \varphi$. ■

Corollary 4.7. *Let $W \subseteq V$ be a k -linear subspace, and i the inclusion map $W \hookrightarrow V$. Then $W \cong \text{Ker}(\text{coker } i)$.*

Proof. Note that $\text{Im } i \cong W$, and apply proposition 4.6. ■

Proposition 4.6 motivates *two* definitions in arbitrary $\mathcal{A}\text{b}$ -categories which are useful in homological algebra.

Definition 4.8. Let $\mathcal{C}$ be an $\mathcal{A}\text{b}$ -category, and $\varphi : x \rightarrow y$ a morphism. The *image* $\text{Im } \varphi$ is defined as $\text{Ker}(\text{coker } \varphi)$. The *coimage* $\text{Coim } \varphi$ is defined as $\text{Coker}(\text{ker } \varphi)$.

Remark 4.9. Recall from definition 4.2 that kernels come with a canonical morphism. This means that there is a morphism $\text{ker coker } \varphi : \text{Ker}(\text{coker } \varphi) \rightarrow y$. We denote this morphism by $\text{im } \varphi : \text{Im } \varphi \rightarrow y$. Similarly, there is a morphism $\text{coim } \varphi = \text{coker ker } \varphi : x \rightarrow \text{Coim } \varphi$.

By corollary 4.7, subspaces of a k -linear space V are precisely kernels in V , which are also precisely images of morphisms into V . Naïvely we might choose to define subobjects in an arbitrary category as kernels or images, but this still isn't quite right. It doesn't really make sense to think of the image of *any* morphism as a subobject. A quotient map, for example, is absolutely not something we would think of as a subobject. We need a categorification of the notion of an *inclusion* map.

It will help to broaden our notion of a subobject. We can think of *any* injective map as a subobject. To see why, let $\varphi : V \rightarrow W$ be a k -linear injective map, and let $i : \text{Im } \varphi \rightarrow W$ be the canonical inclusion. We can then factor φ as

$$\begin{array}{ccccc} & & \varphi & & \\ & \searrow & \curvearrowright & \searrow & \\ V & \xrightarrow{\bar{\varphi}} & \text{Im } \varphi & \xrightarrow{i} & W, \end{array}$$

where $\bar{\varphi}(v) = \varphi(v)$, viewed as an element of $\text{Im } \varphi$. The important thing to note is that $\bar{\varphi}$ is an *isomorphism*. Injective maps can thus be thought of as identifying a subspace $\text{Im } \varphi$ of the codomain, and then including it into the main space. This is why we can think of injections as subobjects. Conversely, we can think of surjections as *quotients*. There are categorical versions of injections and surjections, which we now review.

Definition 4.10. Let $\mathcal{C}$ be a category. A morphism $f : x \rightarrow y$ is said to be *monic* or a *monomorphism* if for any $z \in \mathcal{C}$ and any $\alpha, \beta : z \rightarrow x$,

$$f \circ \alpha = f \circ \beta \implies \alpha = \beta.$$

In other words, if the diagram

$$z \begin{array}{c} \xrightarrow{\alpha} \\ \xrightarrow{\beta} \end{array} x \xrightarrow{f} y$$

commutes¹, then $\alpha = \beta$.

Definition 4.11. Let $\mathcal{C}$ be a category. A morphism $f : x \rightarrow y$ is said to be *epi* or an *epimorphism* if, for any $z \in \mathcal{C}$ and any $\alpha, \beta : y \rightarrow z$,

$$\alpha \circ f = \beta \circ f \implies \alpha = \beta.$$

In other words, if the diagram

$$x \xrightarrow{f} y \begin{array}{c} \xrightarrow{\alpha} \\ \xrightarrow{\beta} \end{array} z$$

commutes, then $\alpha = \beta$.

Proposition 4.12. Let $\varphi : V \rightarrow W$ be a function. Then it is a monomorphism in $\mathbf{Set}$ if and only if it is injective. If it is k -linear and V, W are k -vector spaces, then it is a monomorphism in $\mathbf{Vect}_k$ if and only if it is injective. Dually, if $\psi : V \rightarrow W$ is a surjective map, then it is epi in $\mathbf{Set}$, and if it is k -linear, then it is epi in $k\text{-Vect}$.

Proof. One can show that if φ is injective, then there exists a surjection $\psi : W \rightarrow V$ such that $\psi \circ \varphi = 1_V$. Then if $\varphi \circ \alpha = \varphi \circ \beta$ for any compatible morphisms α, β , then we have

$$\alpha = 1_V \circ \alpha = \psi \circ \varphi \circ \alpha = \psi \circ \varphi \circ \beta = 1_V \circ \beta = \beta.$$

Thus φ is a monomorphism. This proof works in both $\mathbf{Set}$ and $\mathbf{Vect}_k$, but the converse is different for the different categories.

In $\mathbf{Set}$, note that an element $v \in V$ is precisely the same as the morphism $\{*\} \rightarrow V$ given by $* \mapsto v$. Let $v, w : \{*\} \rightarrow V$ be elements of V . Then if $\varphi(v) = \varphi(w)$, then $\varphi \circ v = \varphi \circ w$. If φ is monic, then $v = w$.

In $\mathbf{Vect}_k$, $\{*\}$ is the trivial vector space 0 , and the only morphism $0 \rightarrow V$ is 0 . Thus we need a different proof of the converse. Recall from linear algebra that a k -linear map $\varphi : V \rightarrow W$ is injective if and only if its kernel is trivial. Thus we show that if φ is monic, then $\text{Ker } \varphi \cong 0$. Consider the inclusion $i : \text{Ker } \varphi \hookrightarrow V$ and the zero morphism $0 : \text{Ker } \varphi \rightarrow V$. Since $\varphi \circ i = 0$ and $\varphi \circ 0 = 0$, and since φ is monic, we have $i = 0$. The only way for the inclusion of a subspace to be 0 is if $\text{Ker } \varphi \cong 0$.

The proof that $\text{epi} \iff \text{surjective}$ is left as an exercise; they follow by dualizing the logic for monic $\iff$ injective. ■

Thus, we should think of monics as identifying subobjects in an arbitrary category, and similarly epis should be quotients. However, we would also like *kernels* to be subobjects in view of corollary 4.7, and similarly we would like *cokernels* to be quotients in view of proposition 4.4 and remark 4.5. We should thus ask that monics should be kernels and epis should be cokernels². This motivates conditions (2,3) of the following definition.

Definition 4.13 ([ML98]). An abelian category is an additive category $\mathcal{A}$ such that

1. every morphism has a kernel and a cokernel;

¹It is convention to say that if there are two parallel arrows $z \rightrightarrows x$, then it is not required for them to be equal in order for the diagram to commute.

²Recall that when referring to a morphism as a (co)kernel, we are referring to the canonical injection/projection associated to the (co)kernel object.

2. every monic is a kernel;
3. every epi is a cokernel.

This is the most intuitive formulation of abelian categories, but there is another standard (equivalent) definition which reformulates conditions (2,3) of proposition 4.12. The following proposition motivates the alternative definition.

Proposition 4.14. *Every monic (kernel) in an abelian category $\mathcal{A}$ is the kernel of its cokernel. Dually, every epi (cokernel) in an abelian category is the cokernel of its kernel. Additionally, every morphism f in $\mathcal{A}$ factors as*

$$f = \text{coim } f \circ \text{im } f = \text{coker}(\ker f) \circ \ker(\text{coker } f).$$

Proof. The first two statements are exercises. The third is nontrivial; see [ML98] IX.3 proposition 1. ■

Definition 4.15 ([Wei94]). An abelian category is an additive category $\mathcal{A}$ such that

1. every morphism has a kernel and a cokernel;
2. every monic is the kernel of its cokernel;
3. every epi is the cokernel of its kernel.

We close this section with two propositions that will be useful later on, as well as some important definitions.

Lemma 4.16. *Let $\mathcal{A}$ be an abelian category (more generally, and Ab -category), and $f : x \rightarrow y$. Then $f \circ 0 = 0$ and $0 \circ f = 0$.*

Proof. By definition 2.4, composition distributes over addition, and $0 = 0 + 0$. We have

$$f \circ 0 = f \circ (0 + 0) = (f \circ 0) + (f \circ 0).$$

It follows that

$$(f \circ 0) - (f \circ 0) = (f \circ 0) + (f \circ 0) - (f \circ 0).$$

We thus have $f \circ 0 = 0$. The other statement follows the same. ■

Proposition 4.17. *Let $\mathcal{A}$ be an abelian category. A map $f : x \rightarrow y$ is monic if and only if for all $\varphi : z \rightarrow x$,*

$$f \circ \varphi = 0 \implies \varphi = 0.$$

Dually, f is epi if and only if for all $\varphi : y \rightarrow z$,

$$\varphi \circ f = 0 \implies \varphi = 0.$$

Proof. Let f be monic, and $f \circ \varphi = 0$. We also have that $f \circ 0 = 0$ by lemma 4.16, and since f is monic, by definition 4.10 $\varphi = 0$. Conversely, let f be a morphism such that $f \circ \varphi = 0$ implies $\varphi = 0$. Let $\alpha, \beta : z \rightarrow x$ such that $f \circ \alpha = f \circ \beta$. Then

$$(f \circ \alpha) - (f \circ \beta) = (f \circ \beta) - (f \circ \beta) = 0.$$

By definition 2.4, we have

$$(f \circ \alpha) - (f \circ \beta) = f \circ (\alpha - \beta) = 0.$$

By hypothesis, $\alpha - \beta = 0$, and thus $\alpha = \beta$. Thus, f is monic. The statement for epis follows the same. ■

Definition 4.18. Let $(X_\bullet, d_\bullet)$ be a chain complex of objects in an abelian category. We say $X_\bullet$ is *exact* at i if $\ker d_i = \operatorname{im} d_{i+1}$. There are numerous equivalent characterizations of this definition, and we encourage the reader to find some. One useful recharacterization is that the homology (which will be defined in ?? 5.3) of $X_\bullet$ is 0 at i . We say $X_\bullet$ is *exact* if it is exact in all degrees.

5 Homology

We begin this section with some examples of abelian categories. Clearly, $\mathbf{Vect}_k$ is an abelian category, and so are $\mathcal{A}b$ and $\mathbf{Mod}_R$, since we modeled abelian categories after these categories. The following definition gives another example.

Definition 5.1. Let $\mathcal{A}$ be an abelian category. The category $\mathbf{Ch}(\mathcal{A})$ of *chain complexes* in $\mathcal{A}$ is defined by taking objects to be chain complexes $x_\bullet \in \mathbf{Ch}(\mathcal{A})$, and morphisms $f_\bullet : x_\bullet \rightarrow y_\bullet$ are collections of functions $\{f_n\}_{n \in \mathbb{Z}}$ which commute with differentials in the sense that the diagram

$$\begin{array}{ccccccc} \cdots & \xrightarrow{d_{n+2}} & x_{n+1} & \xrightarrow{d_{n+1}} & x_n & \xrightarrow{d_n} & x_{n-1} \xrightarrow{d_{n-1}} \cdots \\ & & \downarrow f_{n+1} & & \downarrow f_n & & \downarrow f_{n-1} \\ \cdots & \xrightarrow{d_{n+2}} & y_{n+1} & \xrightarrow{d_{n+1}} & y_n & \xrightarrow{d_n} & y_{n-1} \xrightarrow{d_{n-1}} \cdots \end{array}$$

commutes. More explicitly,

$$f_{n-1} \circ d_n = d_n \circ f_n.$$

Composition and identity are given pointwise.

Proposition 5.2. *The category $\mathbf{Ch}(\mathcal{A})$ is abelian.*

The proof of this fact is a good exercise. The categorically-minded reader can simplify the proof by recognizing $\mathbf{Ch}(\mathcal{A})$ as a full subcategory of $\mathbf{Fun}(\mathbb{Z}, \mathcal{A})$, and recalling how (co)limits in functor categories are defined. It then remains only to show that $\mathbf{Ch}(\mathcal{A})$ is stable under the relevant (co)limits.

Another example of an abelian category in particular use in geometry is as follows. Given an abelian category $\mathcal{A}$ and a topological space X (more generally, a site), the category $\mathbf{Shf}_{\mathcal{A}}(X)$ of sheaves on X with values in $\mathcal{A}$ is an abelian category. This means categories such as the category $\mathbf{Shf}_{\mathcal{A}b}(X)$ of sheaves of abelian groups are useful, since one can immediately do homological algebra with sheaves.

Construction 5.3. In order to actually do homological algebra, we now do good on our promise to construct the homology of chain complexes in an abelian group $\mathcal{A}$. Let $x_\bullet$ be a chain complex. Recall that in $\mathbf{Vect}_k$, we defined homology as $H_i(x_\bullet) = \operatorname{Ker} d_i / \operatorname{Im} d_{i+1}$. This can be viewed as the cokernel of the inclusion $\operatorname{Im} d_{i+1} \hookrightarrow \operatorname{Ker} d_i$. We thus need to construct a map $\operatorname{Im} d_{i+1} \rightarrow \operatorname{Ker} d_i$ in an arbitrary abelian category which we can take the cokernel of.

Consider just a part of the chain complex

$$x_{n+1} \xrightarrow{d_{n+1}} x_n \xrightarrow{d_n} x_{n-1}.$$

By proposition 4.14, d_{n+1} decomposes as

$$\begin{array}{ccccc} & & \operatorname{Im} d_{n+1} & & \\ & \nearrow & \downarrow \operatorname{im} d_{n+1} & \searrow & \\ \operatorname{coim} d_{n+1} & \xrightarrow{\quad} & x_{n+1} & \xrightarrow{d_{n+1}} & x_n \xrightarrow{d_n} x_{n-1}. \end{array}$$

Here we have noted a corollary of proposition 4.14 that $\text{Coim } f \cong \text{Im } f$ for any morphism f . Note that $d_n \circ d_{n+1} = 0$, and thus

$$d_n \circ \text{im } d_{n+1} \circ \text{coim } d_{n+1} = 0.$$

Since $\text{coim } d_{n+1}$ is epi, by proposition 4.17 we have that

$$d_n \circ \text{im } d_{n+1} \circ \text{coim } d_{n+1} = 0 \iff d_n \circ \text{im } d_{n+1} = 0.$$

Thus, $\text{im } d_{n+1}$ factors through the kernel

$$\begin{array}{ccccc} & \text{Im } d_{n+1} & \xrightarrow{\psi} & \text{Ker } d_n & \\ & \uparrow \text{coim } d_{n+1} & \searrow & \downarrow \ker d_n & \\ x_{n+1} & \xrightarrow{d_{n+1}} & x_n & \xrightarrow{d_n} & x_{n-1}. \end{array}$$

We then define $H_n(x_\bullet)$ as $\text{Coker}(\psi)$

$$\begin{array}{ccccc} & \text{Im } d_{n+1} & \xrightarrow{\psi} & \text{Ker } d_n & \xrightarrow{\text{coker } \psi} \text{Coker}(\psi) =: H_n(x_\bullet) \\ & \uparrow \text{coim } d_{n+1} & \searrow & \downarrow \ker d_n & \\ x_{n+1} & \xrightarrow{d_{n+1}} & x_n & \xrightarrow{d_n} & x_{n-1}. \end{array}$$

Proposition 5.4. *Let $\mathcal{A}$ be an abelian category. Then for any n , the n th homology is a functor $H_n : \text{Ch}(\mathcal{A}) \rightarrow \mathcal{A}$.*

Proof. We already have an object function $x_\bullet \mapsto H_n(x_\bullet)$, so we need to show that morphisms $f_\bullet : x_\bullet \rightarrow y_\bullet$ induce morphisms $H_n(f_\bullet) : H_n(x_\bullet) \rightarrow H_n(y_\bullet)$ in a functorial manner. First, we construct the morphism $H_n(f_\bullet)$. Consider the diagram

$$\begin{array}{ccccccc} & \text{Im } d_{n+1}^x & \xrightarrow{\psi_x} & \text{Ker } d_n^x & \xrightarrow{\text{coker } \psi_x} & H_n(x_\bullet) & \\ & \uparrow \text{coim } d_{n+1}^x & \searrow \text{im } d_{n+1}^x & \downarrow \ker d_n^x & & & \\ x_{n+1} & \xrightarrow{d_{n+1}^x} & x_n & \xrightarrow{d_n^x} & x_{n-1} & & \\ \downarrow f_{n+1} & & \downarrow f_n & & \downarrow f_{n-1} & & \\ y_{n+1} & \xrightarrow{d_{n+1}^y} & y_n & \xrightarrow{d_n^y} & y_{n-1} & & \\ & \uparrow \text{coim } d_{n+1}^y & \nwarrow \text{im } d_{n+1}^y & \uparrow \ker d_n^y & & & \\ & \text{Im } d_{n+1}^y & \xrightarrow{\psi_y} & \text{Ker } d_n^y & \xrightarrow{\text{coker } \psi_y} & H_n(y_\bullet) & \end{array}$$

By commutativity of the diagram,

$$d_n^y \circ f_n \circ \ker d_n^x = f_{n-1} \circ d_n^x \circ \ker d_n^x = 0.$$

It follows that $f_n \circ \ker d_n^x$ factors uniquely through $\ker d_n^y$ such that the diagram commutes:

$$\begin{array}{ccccc}
& & 0 & & \\
& \swarrow & & \searrow & \\
\text{Ker } d_n^x & \xrightarrow{f_n \circ \ker d_n^x} & y_n & \xrightarrow{d_n^y} & y_{n-1} \\
& \searrow \sigma & \uparrow \ker d_n^y & \nearrow 0 & \\
& & \text{Ker } d_n^y & &
\end{array}$$

This gives an induced map $\sigma : \text{Ker } d_n^x \rightarrow \text{Ker } d_n^y$. By commutativity of the diagram, we also have the commutative diagram of solid arrows

$$\begin{array}{ccccccc}
x_{n+1} & \xrightarrow{\text{coim } d_{n+1}^x} & \text{Im } d_{n+1}^x & \xrightarrow{\psi_x} & \text{Ker } d_n^x & \xrightarrow{\ker d_n^x} & x_n \xrightarrow{f_n} y_n \\
\downarrow f_{n+1} & & & & \downarrow \sigma & & \nearrow \ker d_n^y \\
y_{n+1} & \xrightarrow{\text{coim } d_{n+1}^y} & \text{Im } d_{n+1}^y & \xrightarrow{\psi_y} & \text{Ker } d_n^y & &
\end{array}$$

commutes. We want to show that if we include the dotted arrow σ , then the diagram will still commute. The triangle on the right commutes by definition of σ . The rectangle then follows by commutativity of the border and that $\ker d_n^y$ is monic. Of course, including $H_n(y_\bullet)$ in this diagram gives

$$\begin{array}{ccccccc}
x_{n+1} & \xrightarrow{\text{coim } d_{n+1}^x} & \text{Im } d_{n+1}^x & \xrightarrow{\psi_x} & \text{Ker } d_n^x & \xrightarrow{\ker d_n^x} & x_n \xrightarrow{f_n} y_n \\
\downarrow f_{n+1} & & & & \downarrow \sigma & & \nearrow \ker d_n^y \\
y_{n+1} & \xrightarrow{\text{coim } d_{n+1}^y} & \text{Im } d_{n+1}^y & \xrightarrow{\psi_y} & \text{Ker } d_n^y & \xrightarrow{\text{coker } \psi_y} & H_n(y_\bullet) \\
& & & & & \searrow & \\
& & & & & 0 &
\end{array}$$

We of course then see by lemma 4.16 that

$$\text{coker } \psi_y \circ \psi_y \circ \text{coim } d_{n+1}^y \circ f_{n+1} = 0.$$

By commutativity of the diagram, it follows that

$$\text{coker } \psi_y \circ \sigma \circ \psi_x \circ \text{coim } d_n^x = 0.$$

Since the coimage is epi, by proposition 4.17 we have

$$\text{coker } \psi_y \circ \sigma \circ \psi_x = 0.$$

Thus, $\text{coker } \psi_y \circ \sigma$ factors uniquely through $\text{coker } \psi_x$:

$$\begin{array}{ccccccc}
& & 0 & & & & \\
& \swarrow & & \searrow & & & \\
\text{Im } d_{n+1}^x & \xrightarrow{\psi_x} & \text{Ker } d_n^x & \xrightarrow{\sigma} & \text{Ker } d_n^y & \xrightarrow{\text{coker } \psi_y} & H_n(y_\bullet) \\
& & \downarrow \text{coker } \psi_x & & & \nearrow H_n(f_\bullet) & \\
& & H_n(x_\bullet) & & & &
\end{array}$$

Thus there exists a morphism $H_n(f_\bullet)$, as claimed. We must now show that this definition is functorial.

Let $1_{x_\bullet} : x_\bullet \rightarrow x_\bullet$ be the identity. Then σ is defined as the dashed arrow witnessing the commutativity of the diagram

$$\begin{array}{ccccc}
 & & 0 & & \\
 & \nearrow & & \searrow & \\
 \text{Ker } d_n^x & \xrightarrow{1_{x_n} \circ \ker d_n^x} & x_n & \xrightarrow{d_n^x} & x_{n-1} \\
 & \searrow \sigma & \uparrow \ker d_n^x & \nearrow 0 & \\
 & & \text{Ker } d_n^x & &
 \end{array}$$

In particular, $\ker d_n^x \circ \sigma = \ker d_n^x \circ 1_{\text{Ker } d_n^x}$. Since kernels are monic in abelian categories, $\sigma = 1_{\text{Ker } d_n^x}$. Thus $H_n(1_{x_\bullet})$ arises as the dashed arrow in the diagram

$$\begin{array}{ccccc}
 & & 0 & & \\
 & \nearrow & & \searrow & \\
 \text{Im } d_{n+1}^x & \xrightarrow{\psi_x} & \text{Ker } d_n^x & \xrightarrow{\text{coker } \psi_x} & H_n(x_\bullet) \\
 & & \downarrow \text{coker } \psi_x & \nearrow H_n(1_{x_\bullet}) & \\
 & & H_n(x_\bullet) & &
 \end{array}$$

Note that if the dashed arrow were 1, the diagram would commute. By uniqueness of the induced arrow, $H_n(1_{x_\bullet}) = 1_{H_n(x_\bullet)}$.

Now let $f_\bullet : x_\bullet \rightarrow y_\bullet$ and $g_\bullet : y_\bullet \rightarrow z_\bullet$:

$$\begin{array}{ccccc}
 x_{n+1} & \xrightarrow{d_{n+1}^x} & x_n & \xrightarrow{d_n^x} & x_{n-1} \\
 f_{n+1} \downarrow & & \downarrow f_n & & \downarrow f_{n-1} \\
 y_{n+1} & \xrightarrow{d_{n+1}^y} & y_n & \xrightarrow{d_n^y} & y_{n-1} \\
 g_{n+1} \downarrow & & \downarrow g_n & & \downarrow g_{n-1} \\
 z_{n+1} & \xrightarrow{d_{n+1}^z} & z_n & \xrightarrow{d_n^z} & z_{n-1}
 \end{array}$$

a

Remark 5.5. We can in fact show more. If $\mathcal{C}, \mathcal{D}$ are additive categories, a functor $F : \mathcal{C} \rightarrow \mathcal{D}$ is *additive* if it preserves addition, meaning if $f, g : x \rightarrow y$ in $\mathcal{C}$, then $F(f+g) = F(f) + F(g)$. Homology is indeed an *additive* functor, but as the reader can likely guess from the proof of proposition 5.4, proving this result is fairly involved and unenlightening. The reader is encouraged to prove this on their own should they feel so inclined.

Corollary 5.6. *Let $\mathcal{A}$ be an abelian category. Then homology defines a functor $H_\bullet : \text{Ch}(\mathcal{A}) \rightarrow \text{Ch}(\mathcal{A})$.*

Proof. Define the complex $H_\bullet(x_\bullet)$ as the complex whose object in the n th degree is the n th homology $H_n(x_\bullet)$, and whose boundaries are all 0:

$$H_\bullet(x_\bullet) = \cdots \xrightarrow{0} H_{n+1}(x_\bullet) \xrightarrow{0} H_n(x_\bullet) \xrightarrow{0} H_{n-1}(x_\bullet) \xrightarrow{0} \cdots$$

This is clearly a chain complex. Additionally, by lemma 4.16 we see that $H_\bullet(f_\bullet)$ is indeed a morphism $H_\bullet(x_\bullet) \rightarrow H_\bullet(y_\bullet)$ for each $f_\bullet : x_\bullet \rightarrow y_\bullet$; commutativity with the differentials is immediate by lemma 4.16, since they are all 0.

Remark 5.7. It may not be immediately clear what the significance of corollary 5.6 is. The fact that $H_\bullet$ is an endofunctor is interesting, but it may not be obvious why the resulting chain complex is interesting, especially since the boundary morphisms are all 0. To see why we care, consider two arbitrary chain complexes $x_\bullet, y_\bullet$ whose boundary morphisms are 0, and a morphism $f_\bullet : x_\bullet \rightarrow y_\bullet$:

$$\begin{array}{ccccccc} \cdots & \xrightarrow{0} & x_{n+1} & \xrightarrow{0} & x_n & \xrightarrow{0} & x_{n-1} & \xrightarrow{0} & \cdots \\ & & \downarrow f_{n+1} & & \downarrow f_n & & \downarrow f_{n-1} & & \\ \cdots & \xrightarrow{0} & y_{n+1} & \xrightarrow{0} & y_n & \xrightarrow{0} & y_{n-1} & \xrightarrow{0} & \cdots \end{array}$$

By definition 5.1, the morphism $f_\bullet$ can be any collection of functions satisfying the commutativity condition

$$d_{i-1}^y \circ f_i = f_{i-1} \circ d_i^x.$$

However, since the boundary morphisms are all 0, we have

$$0 = 0 \circ f_i = f_{i-1} \circ 0 = 0.$$

Thus, *any* collection of morphisms $f_\bullet = \{f_i : x_i \rightarrow y_i\}_{i \in \mathbb{Z}}$ suffices as a morphism $f_\bullet : x_\bullet \rightarrow y_\bullet$ between chain complexes with 0 boundary.

This tells us something interesting about isomorphisms of chain complexes with 0 boundary. An isomorphism $f_\bullet$ is a morphism of chain complexes is a morphism such that each f_i has an inverse, and the inverses assemble into a morphism $f_\bullet^{-1}$ of chain complexes. However, since the commutativity condition for chain complexes with 0 boundary is automatically satisfied, *any* such collection of *isomorphisms* $f_\bullet$ is an isomorphism of chain complexes, provided the complexes have 0 boundary.

Now we can apply this to the functor of corollary 5.6. Suppose we have two chain complexes $x_\bullet, y_\bullet$. Since the homology complexes $H_\bullet(x_\bullet), H_\bullet(y_\bullet)$ have 0 boundary, an isomorphism $f_\bullet : H_\bullet(x_\bullet) \rightarrow H_\bullet(y_\bullet)$ is simply a collection of isomorphisms $f_i : H_i(x_\bullet) \cong H_i(y_\bullet)$. In other words, $H_\bullet(x_\bullet) \cong H_\bullet(y_\bullet)$ if and only if $x_\bullet$ and $y_\bullet$ have the same homology in each degree. The real power of this fact comes from the following idea. Suppose we have some morphism $f_\bullet : x_\bullet \rightarrow y_\bullet$ in $\text{Ch}(\mathcal{A})$ such that $H_\bullet(f_\bullet)$ is an *isomorphism* $H_\bullet(x_\bullet) \cong H_\bullet(y_\bullet)$. Not only do we now know that $x_\bullet$ and $y_\bullet$ have the same homology, but we have a *regular morphism* $f_\bullet$ which tells us this fact. Morphisms of this form are important, and deserve a name.

Definition 5.8. Let $\mathcal{A}$ be an abelian category. A *quasi-isomorphism* is a morphism $f_\bullet : x_\bullet \rightarrow y_\bullet$ of chain complexes which lowers to an isomorphism in homology $H_\bullet(f_\bullet) : H_\bullet(x_\bullet) \cong H_\bullet(y_\bullet)$.

Quasi-isomorphisms are extremely significant. Throughout the Colloquium, we have been far more interested in the *homology* of a chain complex, rather than the chain complex itself. Homology is a convenient yet powerful way to extract information from chain complexes, and by proposition 5.4 it is an invariant. Because of this, much of homological algebra is the study of chain complexes *up to homology*. In other words, we would like to consider two chain complexes to be “the same” if they have the same homology. However, quasi-isomorphisms *need not be isomorphisms*, and in particular if two chain complexes have the same homology, *this does not mean they are isomorphic*. This is very different from much of math, where we are often interested in studying objects up to isomorphism, and thus presents some obstacles.

Digression 5.9. It would be nice if we could somehow turn quasi-isomorphisms into isomorphisms. The study of homological algebra would then reduce to the study of chain complexes up to this new notion of isomorphism, and we are already quite good at studying objects up to isomorphism. This is

a very powerful perspective, and a decent amount of modern homological algebra involves *homotopy theory*. Oftentimes, a category $\mathcal{C}$ comes with a class of maps $\mathcal{W}$ which we call *weak equivalences*, and we want those weak equivalences to be isomorphisms. Good examples are quasi-isomorphisms of chain complexes, and (weak) homotopy equivalences of topological spaces. We have a construction called *localization* which can construct a new category wherein the weak equivalences become isomorphisms, but unfortunately this new category can be very poorly behaved and very difficult to work in (for example, it may not be locally small). Homotopy theory can be roughly described as trying to fix these issues by giving $\mathcal{C}$ extra structure, designed to make the weak equivalences behave better when localized. Many important constructions in homological algebra, such as the derived category, the derived ∞ -category, derived functors, and more can be seen as following from homotopical methods.

6 Resolutions

Generally when we study objects “up to isomorphism”, the goal is to replace complex objects with “nice” objects that we already know a lot about, and we do thus by finding an isomorphism between them. Although we are not studying chain complexes up to isomorphism, we should still take motivation from other areas of math. Our goal for this section will be to replace chain complexes with quasi-isomorphic “nice” chain complexes. First we obviously need to know what a “nice” chain complex is.

Definition 6.1. Let $\mathcal{A}$ be an abelian category. An object $p \in \mathcal{A}$ is *projective* if for any epimorphism $f : x \rightarrow y$ and any morphism $g : p \rightarrow y$, there is a dotted arrow as indicated

$$\begin{array}{ccc} & p & \\ \nearrow \gamma & \downarrow g & \\ x & \xrightarrow{f} & y \end{array}$$

rendering the diagram commutative.

Remark 6.2. There is a reformulation of this definition which will not be important for our discussion, but hints at a cofibrantly generated model structure on chain complexes. Definition 6.1 is equivalent to saying that for any diagram of solid arrows where f is epi

$$\begin{array}{ccc} 0 & \xrightarrow{0} & x \\ \downarrow 0 & \nearrow \gamma & \downarrow f \\ p & \xrightarrow{g} & y \end{array}$$

there is a lift as indicated by the dotted arrow, rendering the diagram commutative. In other words, $0 \rightarrow p$ has the left lifting property with respect to epimorphisms.

Projective objects are “nice” because they play nicely with epimorphisms, which as we know from definition 4.13 are precisely the cokernels of $\mathcal{A}$. Projective objects give us ways to factor cokernels, which is useful in many proofs, and many statements are more easily proven for projective objects than general objects. For example, we will see that additive functors represented by projectives $\mathcal{A}(p, -) : \mathcal{A} \rightarrow \mathcal{A}b$ are *exact*. Dual to projectives are *injectives*:

Definition 6.3. Let $\mathcal{A}$ be an abelian category. An object $i \in \mathcal{A}$ is *injective* if for any monomorphism $f : x \rightarrow y$ and any map $g : x \rightarrow i$, there is a dotted arrow as indicated

$$\begin{array}{ccc} x & \xrightarrow{f} & y \\ g \downarrow & \nearrow \gamma & \\ i & & \end{array}$$

rendering the diagram commutative.

The following definition is convenient, since it is a fairly common condition that ensures various important constructions always exist.

Definition 6.4. Let $\mathcal{A}$ be an abelian category. We say $\mathcal{A}$ has *enough projectives* if for all $x \in \mathcal{A}$, there is a projective object p and an epimorphism $p \rightarrow x$. Similarly, we say $\mathcal{A}$ has *enough injectives* if for all $x \in \mathcal{A}$, there is an injective object i and a monomorphism $x \rightarrow i$.

Proposition 6.5. Let $\mathcal{A}$ be an abelian category with enough projectives (resp. injectives). Then $\text{Ch}(\mathcal{A})$ has enough projectives (resp. injectives).

Proof. See [FIND REFERENCE]. ■

Definition 6.6. Let $\mathcal{A}$ be an abelian category. A chain complex $p_\bullet$ in $\text{Ch}(\mathcal{A})$ is a *complex of projectives* if each p_n is projective. A chain complex $i_\bullet$ is a *chain complex of injectives* if each i_n is injective.

Warning 6.7. A chain complex of projectives *does not need to be a projective object* in $\text{Ch}(\mathcal{A})$. Similarly, a chain complex of injectives does not need to be an injective object in $\text{Ch}(\mathcal{A})$.

Although complexes of projectives and injectives need not be projective or injective in $\text{Ch}(\mathcal{A})$, they are still of significant importance due to the fact that they are constructed out of well-behaved objects. The significance of these constructions is illuminated by the notion of a *resolution*.

Definition 6.8. Let $\mathcal{A}$ be an abelian category, and $X \in \mathcal{A}$. Define the chain complex $\iota(X)_\bullet \in \text{Ch}(\mathcal{A})$ as the complex

$$\cdots \xrightarrow{0} 0 \xrightarrow{0} X \xrightarrow{0} 0 \xrightarrow{0} \cdots$$

In particular, we take X to be the 0th degree object, and all other objects are 0.

Definition 6.9. Let $\mathcal{A}$ be an abelian category, and $X \in \mathcal{A}$. A *projective resolution* is a chain complex $P_\bullet$ of projectives with $P_n = 0$ for all $n < 0$, together with a quasi-isomorphism $\epsilon : P_\bullet \rightarrow \iota(X)_\bullet$. An *injective resolution* is a cochain complex of injectives $I^\bullet$ with $I^n = 0$ for all $n < 0$, together with a quasi-isomorphism $\eta : \iota(X)^\bullet \rightarrow I^\bullet$.

Proposition 6.10. Let $\epsilon : P_\bullet \rightarrow \iota(X)_\bullet$ be a projective resolution of $X \in \mathcal{A}$. Then the complex

$$\cdots \longrightarrow P_2 \longrightarrow P_1 \longrightarrow P_0 \xrightarrow{\epsilon_0} X \longrightarrow 0$$

is exact. Dually, if $\eta : \iota(X)^\bullet \rightarrow I^\bullet$ is an injective resolution of $X \in \mathcal{A}$, then the complex

$$0 \longrightarrow X \xrightarrow{\eta_0} I^0 \longrightarrow I^1 \longrightarrow I^2 \longrightarrow \cdots$$

is exact.

Proof. Exercise. Injectives follow by duality. It suffices to show the homology is vanishing at all degrees. Of course, the homology in $P_\bullet$ vanishes for all $n \neq 0$, since it vanishes for $\iota(X)_\bullet$ at all $n \neq 0$, and $P_\bullet$ is quasi-isomorphic to $\iota(X)_\bullet$. We also have that the homology of $P_1 \rightarrow P_0 \rightarrow 0$ is the same as $0 \rightarrow X \rightarrow 0$. The reader should be able to complete the proof without much difficulty. ■

Let us discuss what we have achieved so far. Given an object $X \in \mathcal{A}$, we have found a way to view that object as a chain complex via definition 6.8. *Projective and injective resolutions* of this object can be thought of as special cases of our idea of replacing complexes with “nice” complexes. This special case turns out to be of extreme importance, and we will now illustrate why.

7 Derived functors

One of the numerous reasons that functors are useful in math is because they preserve isomorphisms. Since we are usually interested in studying objects up to isomorphism, the existence of a functor allows us to move isomorphisms from one category to another. However, recall that we are interested in studying chain complexes up to *quasi-isomorphism*, so it’s natural to ask how functors interact with quasi-isomorphisms. We will specialize our discussion to *additive* functors, since it is fairly important that functors preserve 0, which additive functors do. It will be useful to lead with the following construction.

Construction 7.1. Let $F : \mathcal{A} \rightarrow \mathcal{B}$ be an additive functor between abelian categories. Then we may define the functor $\text{Ch}(F) : \text{Ch}(\mathcal{A}) \rightarrow \text{Ch}(\mathcal{B})$ as follows. Given a chain complex $C_\bullet$ in $\text{Ch}(\mathcal{A})$, define $\text{Ch}(F)(C_\bullet)$ as the chain complex

$$\dots \xrightarrow{F(d_{n+2})} F(C)_{n+1} \xrightarrow{F(d_{n+1})} F(C)_n \xrightarrow{F(d_n)} F(C)_{n-1} \xrightarrow{F(d_{n-1})} \dots$$

Note that this is indeed a chain complex, since additive functors induce *abelian group homomorphisms* $\mathcal{A}(C_i, C_{i-1}) \rightarrow \mathcal{B}(F(C_i), F(C_{i-1}))$. In particular, they preserve the zero morphism, meaning that

$$F(d_{i-1}) \circ F(d_i) = F(d_{i-1} \circ d_i) = F(0) = 0.$$

To define the action of $\text{Ch}(F)$ on morphisms, let $f_\bullet : C_\bullet \rightarrow D_\bullet$ be a morphism of chain complexes in $\text{Ch}(\mathcal{A})$. Then by functoriality of F , the diagram

$$\begin{array}{ccccccc} \dots & \xrightarrow{F(d_{n+2})} & F(C_{n+1}) & \xrightarrow{F(d_{n+1})} & F(C_n) & \xrightarrow{F(d_n)} & F(C_{n-1}) \xrightarrow{F(d_{n-1})} \dots \\ & & \downarrow F(f_{n+1}) & & \downarrow F(f_n) & & \downarrow F(f_{n-1}) \\ \dots & \xrightarrow{F(d_{n+2})} & F(C_{n+1}) & \xrightarrow{F(d_{n+1})} & F(C_n) & \xrightarrow{F(d_n)} & F(C_{n-1}) \xrightarrow{F(d_{n-1})} \dots \end{array}$$

commutes. Thus, $\text{Ch}(F)(f_\bullet)$ is a morphism of chain complexes $\text{Ch}(F)(C_\bullet) \rightarrow \text{Ch}(F)(D_\bullet)$.

This is an important construction that will appear periodically. For now, we turn our attention back to quasi-isomorphisms. Unfortunately, additive functors need not preserve quasi-isomorphisms in general, but there are important classes of functors that do.

Definition 7.2. Let $\mathcal{A}$ be an abelian category. A *short exact sequence* in $\text{Ch}(\mathcal{A})$ is an exact complex of the form

$$0 \xrightarrow{0} A \xrightarrow{\varphi} B \xrightarrow{\psi} C \xrightarrow{0} 0.$$

Definition 7.3. Let $F : \mathcal{A} \rightarrow \mathcal{B}$ be an additive functor between abelian categories. We say F is *exact* if $\text{Ch}(F)$ takes short exact sequences to short exact sequences. More concretely, if

$$0 \xrightarrow{0} A \xrightarrow{\varphi} B \xrightarrow{\psi} C \xrightarrow{0} 0$$

is a short exact sequence in $\text{Ch}(\mathcal{A})$, then

$$0 \xrightarrow{0} F(A) \xrightarrow{F(\varphi)} F(B) \xrightarrow{F(\psi)} F(C) \xrightarrow{0} 0$$

is a short exact sequence in $\text{Ch}(\mathcal{B})$.

Proposition 7.4. Let $F : \mathcal{A} \rightarrow \mathcal{B}$ be an additive functor. If F is exact, then $\text{Ch}(F)$ preserves quasi-isomorphisms.

Proof. If F is exact, then it can be shown that it preserves finite limits and colimits. It thus preserves homology, which is equivalent to saying it *commutes* with homology. If $H_\bullet(f_\bullet)$ is an isomorphism, then $H_\bullet(\text{Ch}(F)(f_\bullet) = \text{Ch}(F)(H_\bullet(f_\bullet))$ is an isomorphism since functors preserve isomorphisms. ■

Most functors are in general not exact. However, there are two *much* more common conditions that can be thought of as “almost exact”.

Definition 7.5. Let $F : \mathcal{A} \rightarrow \mathcal{B}$ be an additive functor between abelian categories. We say F is *left exact* if for any short exact sequence

$$0 \xrightarrow{0} A \xrightarrow{\varphi} B \xrightarrow{\psi} C \xrightarrow{0} 0,$$

the sequence

$$0 \xrightarrow{0} F(A) \xrightarrow{F(\varphi)} F(B) \xrightarrow{F(\psi)} F(C)$$

is exact. Similarly, we say F is *right exact* if the sequence

$$F(A) \xrightarrow{F(\varphi)} F(B) \xrightarrow{F(\psi)} F(C) \xrightarrow{0} 0$$

is exact.

Left and right exact functors abound in homological algebra. For example, if $F \dashv G$ is an adjunction, then F is right exact and G is left exact. This is a special case of the following proposition:

Proposition 7.6. Let $F : \mathcal{A} \rightarrow \mathcal{B}$ be an additive functor between abelian categories. Then F is left exact if it preserves finite limits. Dually, F is right exact if it preserves finite colimits.

Proof. Kernels are (finite) limits and cokernels are (finite) colimits. It thus suffices by definition 4.18 to show that additive functors which preserve finite limits must preserve monics, and the other statement follows by duality. Indeed, by definition 4.13, monics are kernels, which are (finite) limits, which are preserved by F . ■

Not many functors preserve both finite limits and colimits, but quite a few preserve at least one of those. For example, we have

Corollary 7.7. Let $\mathcal{A}$ be an abelian category and $X \in \mathcal{A}$. Then the representable functor $\mathcal{A}(X, -) : \mathcal{A} \rightarrow \mathcal{Ab}$ is left exact, and $\mathcal{A}(-, X) : \mathcal{A}^{\text{op}} \rightarrow \mathcal{Ab}$ is right exact.

Corollary 7.8. *Let V be a k -vector space (or R -module). Then $- \otimes_k V$ is right exact.*

Proof. $- \otimes_k V \dashv k\text{-Vect}(V, -)$. ■

Remark 7.9. An R -module M is *flat* if $- \otimes_R M$ is exact. For those unfamiliar with R -modules, vector spaces are a special case of R -modules. However, all k -vector spaces are projective, and all projective modules turn out to be flat. Thus, all k -vector spaces are flat.

Since left and right exact functors are *almost* exact, it would be nice if we could turn them into exact functors somehow. This yields the notion of a *derived functor*. We will take our first step towards their construction with the notion of a *point-set derived functor*. These do not exist in general, and we must restrict to the category $\text{Ch}^-(\mathcal{A})$.

Definition 7.10. Let $\mathcal{A}$ be an abelian category. A chain complex $C_\bullet$ is *bounded above* if $C_n = 0$ for all $n > 0$, and it is *bounded below* if $C_n = 0$ for all $n < 0$. We write $\text{Ch}^+(\mathcal{A})$ for the full subcategory of $\text{Ch}(\mathcal{A})$ spanned by chain complexes which are bounded *below* (the $+$ indicates they are concentrated in positive degrees), and $\text{Ch}^-(\mathcal{A})$ for the category of chain complexes bounded *above*.

Note that if $F : \mathcal{A} \rightarrow \mathcal{B}$ is additive, then there are functors $\text{Ch}^\pm(F) : \text{Ch}^\pm(\mathcal{A}) \rightarrow \text{Ch}^\pm(\mathcal{B})$ defined analogously to ?? 7.1.

Construction 7.11. Let $F : \mathcal{A} \rightarrow \mathcal{B}$ be right exact, and let $\mathcal{A}$ have enough projectives. We will define a functor $\mathbb{L}F : \text{Ch}^-(\mathcal{A}) \rightarrow \text{Ch}^-(\mathcal{B})$ as follows. Let $C_\bullet$ be a chain complex in $\text{Ch}^-(\mathcal{A})$, and let $P_\bullet$ be a complex of projectives with a quasi-isomorphism $\epsilon : P_\bullet \rightarrow C_\bullet$. It is a (nontrivial) fact that such a complex can always be chosen in a functorial way. We will discuss this more in-depth later for a special case. For now, we define $\mathbb{L}F(C_\bullet) = \text{Ch}^-(F)(P_\bullet)$. The action on morphisms is not obvious. Given $f_\bullet : C_\bullet \rightarrow D_\bullet$, together with $\epsilon_C : P_\bullet^C \rightarrow C_\bullet$ and $\epsilon_D : P_\bullet^D \rightarrow D_\bullet$, we obtain $\text{Ch}^-(F)(f_\bullet \circ \epsilon_C) : P_\bullet^C \rightarrow D_\bullet$, but it is not clear how to get a morphism $D_\bullet \rightarrow P_\bullet^D$. It is yet another nontrivial fact that we can construct one in a functorial way. We thus obtain a functor $\mathbb{L}F : \text{Ch}^-(\mathcal{A}) \rightarrow \text{Ch}^-(\mathcal{B})$.

We understand that we have asked the reader to take us at our word for these two nontrivial claims. We are mainly using the construction $\mathbb{L}F$ as *motivation* for a construction which we will shortly see, and we will prove these statements in the context of this new construction when we see it. For now, we ask the reader to take us at our word on one more thing: the functor $\mathbb{L}F$ preserves quasi-isomorphisms, *and* is universal in the sense that there is a natural transformation $q : \mathbb{L}F \Rightarrow \text{Ch}^-(F)$ such that for any other functor G which preserves quasi-isomorphisms and has a natural transformation $\alpha : G \Rightarrow \text{Ch}^-(F)$, α factors uniquely through q . In homotopy theory, $\mathbb{L}F$ would simply be called the derived functor of F , but in homological algebra that term is reserved for a slightly different (but related) construction. We will instead call $\mathbb{L}F$ the *point-set left derived functor* of F . This terminology is taken from [Shu09] and is generally considered nonstandard.

We have now taken a right exact functor $F : \mathcal{A} \rightarrow \mathcal{B}$, and constructed a functor which preserves quasi-isomorphisms $\mathbb{L}F : \text{Ch}^-(\mathcal{A}) \rightarrow \text{Ch}^-(\mathcal{B})$. It would be nice if we could lower it to a functor $\mathcal{A} \rightarrow \mathcal{B}$ which is exact. We in fact have multiple ways of accomplishing this.

Construction 7.12. Let $F : \mathcal{A} \rightarrow \mathcal{B}$ be right exact, and let $\mathcal{A}$ have enough projectives. We can include $\mathcal{A}$ into $\text{Ch}(\mathcal{A})$ in a canonical (functorial) way using the map $\iota : \mathcal{A} \rightarrow \text{Ch}(\mathcal{A})$. Recall that given $X \in \mathcal{A}$, we defined the complex $\iota(X)_\bullet$ as the complex

$$\dots \xrightarrow{0} 0 \xrightarrow{0} X \xrightarrow{0} 0 \xrightarrow{0} \dots,$$

where the X was placed in the 0th degree. It turns out that we can take projective resolutions in a functorial way, so we could imagine defining $\mathbb{L}F$ on $\text{Im } \iota$, yielding a functor $\mathbb{L}F \circ \iota : \mathcal{A} \rightarrow \text{Ch}(\mathcal{B})$. Note here that in ?? 7.11, we replaced our complex with a quasi-isomorphic complex of projectives. Recall from definition 6.9 that such a replacement for $\iota(X)_\bullet$ is a *projective resolution*. Thus, $\mathbb{L}F \circ \iota$ can be described as taking a projective resolution of X , then mapping under $\text{Ch}(F)$. We now only need a functor $\text{Ch}(\mathcal{B}) \rightarrow \mathcal{B}$. The most natural choice is a homology functor $H_i : \text{Ch}(\mathcal{B}) \rightarrow \mathcal{B}$. We can in fact choose *any* homology functor H_i for any i . Define L_iF as the functor $H_i \circ \mathbb{L}F \circ \iota$. Computing $L_iF(X)$ can thus be described as the following process:

1. take a projective resolution $\epsilon : P_\bullet \rightarrow X$,
2. compute $\text{Ch}(F)(P_\bullet)$,
3. and take the i th homology of the resulting complex.

We call L_iF the i th *left derived functor* of F .

Remark 7.13. My method of recovering L_iF from $\mathbb{L}F$ is not very standard. It is technically mostly rigorous, since the process of taking a projective resolution for a complex can be viewed as a functor $Q : \text{Ch}(\mathcal{A}) \rightarrow \text{Ch}(\mathcal{A})$, and thus restricting along ι yields a deformation $Q|_{\text{Im } \iota}$ for F if we forget about the requirement that deformations be endofunctors. We then obtain $\mathbb{L}F$ in the claimed way by taking F on a deformation retract of $Q|_{\text{Im } \iota}$.

However, it is more common to simply forget about all this and work in the derived category $\mathcal{D}(\mathcal{A})$. We unfortunately did not have time to develop derived categories or Cartan-Eilenberg resolutions, so I used a less advanced and less standard approach since. For more on the standard approach, see [Wei94] chapter 10.

Remark 7.14. Derived functors can be defined in much more generality. Given *any* functor $F : \mathcal{A} \rightarrow \mathcal{B}$, we can define L_iF by taking a projective resolution of $X \in \mathcal{A}$, then computing $H_i \circ \text{Ch}(F)$ on that resolution. We focused our discussion on left and right exact functors because this derivation process turns those functors into *exact* functors, as we will see.

We have three important questions we would like to answer.

- Can we prove that L_iF is invariant (up to isomorphism) under choice of projective resolution? Moreover, can we prove that such a resolution always exists, and is functorial?
- Is L_iF exact?
- Can the derived functors be characterized by some universal property?

We will consider the first problem in the remainder of this section, and we delegate the final two to section 8, which tie back into classical homological algebra.

Theorem 7.15 ([Wei94]). *Let $\epsilon_\bullet : P_\bullet \rightarrow \iota(X)_\bullet$ be a projective resolution of $X \in \mathcal{A}$ and $f : X \rightarrow Y$ a morphism in $\mathcal{A}$. Then for any projective resolution $\eta : Q_\bullet \rightarrow \iota(Y)_\bullet$ of Y , there is a morphism of chain complexes $f_\bullet : P_\bullet \rightarrow Q_\bullet$ lifting f in the sense that $\eta_0 \circ f_0 = f \circ \epsilon_0$. The morphism $f_\bullet$ is unique up to chain homotopy.*

Proof. See [Wei94, 2.2] theorem 2.2.6. ■

Lemma 7.16. *Let $\mathcal{A}$ be an abelian category with enough projectives. Then each $X \in \mathcal{A}$ has a projective resolution.*

Proof. We follow [Wei94, 2.2] lemma 2.2.5 for this part. Since $\mathcal{A}$ has enough projectives, there is an epimorphism $\varepsilon_0 : P_0 \rightarrow X$ for some P_0 . Set $M_0 = \text{Ker } \varepsilon_0$, and continue inductively. Given M_{n-1} , there is an epimorphism $\varepsilon_n : P_n \rightarrow M_{n-1}$ with P_n projective; set $M_n = \text{Ker } \varepsilon_n$. The complex $P_\bullet$ is a projective resolution of X . ■

Lemma 7.17. *Let $\mathcal{A}$ be an abelian category, and $P_\bullet, Q_\bullet$ two projective resolutions for some $X \in \mathcal{A}$. Then there are quasi-isomorphisms $f_\bullet : P_\bullet \rightrightarrows Q_\bullet : g$.*

Proof. Let $1_X : X \rightarrow X$. Consider the first X with the resolution $P_\bullet$ and the second with the resolution $Q_\bullet$. Thus, 1_X lifts to some $\varepsilon_\bullet : P_\bullet \rightarrow Q_\bullet$. If we flip the direction before lifting, we get a map $\eta_\bullet : Q_\bullet \rightarrow P_\bullet$. Composing gives a map $\eta_\bullet \circ \varepsilon_\bullet : P_\bullet \rightarrow P_\bullet$ which is a lift of $1_X \circ 1_X = 1_X$, but $1_{P_\bullet}$ is also a lift of 1_X . By theorem 7.15, $1_{P_\bullet}$ and $\eta_\bullet \circ \varepsilon_\bullet$ are chain homotopic, and thus $\eta_\bullet, \varepsilon_\bullet$ are a chain homotopy equivalence.

Chain homotopy equivalences are a special case of quasi-isomorphisms. For the proof of this fact, see [Wei94, 1.4] lemma 1.4.5 which states that chain homotopic maps induce the same maps on homology. This fact then follows as a corollary: the fact that $\eta_\bullet, \varepsilon_\bullet$ are chain homotopy equivalences means that $\eta_\bullet \circ \varepsilon_\bullet$ and $\varepsilon_\bullet \circ \eta_\bullet$ are chain homotopic to the identity, meaning they induce the identity under homology. By functoriality of homology, $\eta_\bullet, \varepsilon_\bullet$ are inverses in homology, and thus quasi-isomorphisms. ■

Theorem 7.18. *Let $F : \mathcal{A} \rightarrow \mathcal{B}$ be an additive, right exact functor between abelian categories, and let $\mathcal{A}$ have enough projectives. Then $L_i F$ exists for all i , and is unique up to isomorphism.*

Proof. By lemma 7.16, there is a replacement function $Q : \mathcal{A} \rightarrow \text{Ch}(\mathcal{A})$ mapping objects to projective resolutions. By ?? 7.1, there is a functor $\text{Ch}(F) : \text{Ch}(\mathcal{A}) \rightarrow \text{Ch}(\mathcal{B})$, and by proposition 5.4 homology is functorial $H_i : \text{Ch}(\mathcal{B}) \rightarrow \mathcal{B}$. By lemma 7.17, each choice of a projective resolution is quasi-isomorphic, and since H_i identifies quasi-isomorphic complexes up to isomorphism, the map is well-defined. Continuing, any choice of lift is chain homotopic, and homology identifies chain homotopic maps ([Wei94, 1.4] lemma 1.4.5). Thus, the morphism function is well-defined, giving a functor. Since $\text{Ch}(F), H$ are additive, the functor is easily shown to be additive. Uniqueness up to isomorphism follows by lemma 7.17. ■

By the usual abstract nonsense, left derived functors have the dual notion of a *right derived functor*. Moreover, all of the above statements dualize to the dual notion, meaning we automatically obtain that right derived functors are unique up to isomorphism, exact, additive, and all that stuff. We also obtain some facts about injective resolutions. We now detail the construction of right derived functors.

Construction 7.19. Let $F : \mathcal{A} \rightarrow \mathcal{B}$ be a *left exact* functor, and let $\mathcal{A}$ have enough injectives. Because we are dualizing left derived functors, we must work with the opposite category $\text{Ch}(\mathcal{A})^{\text{op}}$. However, this is just the category of *cochain complexes*, which works the exact same way. We prefer to simply denote this category by $\text{Ch}(\mathcal{A})$ due to this fact, with the understanding that this now refers to cochain complexes. We define the map $\text{Ch}(F) : \text{Ch}(\mathcal{A}) \rightarrow \text{Ch}(\mathcal{B})$ by following ?? 7.1 for cochain complexes. *right derived functors* $R^i F$ are computed as follows:

1. Take an injective resolution $\eta^\bullet : \iota(X)^\bullet \rightarrow I^\bullet$ of $X \in \mathcal{A}$,
2. compute $\text{Ch}(F)(I^\bullet)$,
3. and take the i th cohomology $H^i \text{Ch}(F)(I^\bullet)$.

8 δ -functors

In this section, we will discuss the relation of derived functors to exact sequences, and show that they are universal δ -functors, cementing their usefulness in homological algebra. We first return to a question we asked in section 7: are derived functors exact? The answer turns out to be no, and this turns out to be fairly simple to see.

Proposition 8.1. *Let $F : \mathcal{A} \rightarrow \mathcal{B}$ be right exact. Then $L_0F \cong F$.*

Proof. Let $X \in \mathcal{A}$. Choose a projective resolution $\varepsilon_\bullet : P_\bullet \rightarrow \iota(X)_\bullet$. Then by construction, $L_0F(X) = (H_0 \circ \text{Ch}(F))(P_\bullet)$. Note that

$$F(P_1) \longrightarrow F(P_0) \longrightarrow F(X) \longrightarrow 0$$

is exact, since F is right exact and the sequence is exact before applying F by hypothesis. Thus, $(H_0 \text{Ch}(F))(P_\bullet) = (H_0F)(P_0) \cong F(X)$ by exactness. ■

In particular, if F is not exact, then L_0F is not. To summarize, we were able to take a right exact functor F and construct a new functor $\mathbb{L}F$ which preserves quasi-isomorphisms. We lowered this to $\mathcal{A} \rightarrow \mathcal{B}$, but the resulting functors are not in general exact, and do not generally preserve quasi-isomorphisms either. The question then becomes: *what are derived functors?*

Recall that derived functors were constructed by essentially replacing objects $X \in \mathcal{A}$ with “nice” complexes of projectives, and then computing the homology. Homology generally measures the failure of a complex to be exact, so what if derived functors measure the failure of a *functor* to be exact?

Definition 8.2. By corollary 7.8, given R a ring and M an S -module, $- \otimes_S M$ is right exact and $S\text{-Mod}(M, -)$ is left exact. Define the functor $\text{Tor}_i^S(-, M)$ as the left derived functor $L_i(- \otimes_R M)$, and define $\text{Ext}_S^i(M, -)$ as the right derived functor $R^i S\text{-Mod}(M, -)$.

9 Homotopy

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